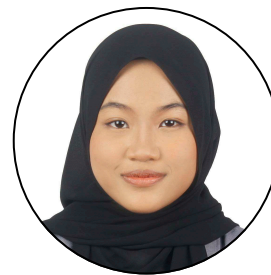


NURUL KHAIRIN BINTI ZULKARNAIN



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📍 No. 34, Jalan PUJ 1/4, Taman Puncak Jalil

Career Objective

As a recent graduate in Chemical Engineering Technology, I am eager to transition into a dynamic role where I can leverage my technical skills to drive operational improvements, enhance product quality, and provide innovative solutions. Whether in consulting, project management, or process optimization, I am ready to apply my problem-solving capabilities to tackle complex challenges in diverse industries.

Education

- **UNIVERSITY COLLEGE TATI (UC TATI)** 2021 - 2025
 - Bachelor of Chemical Engineering Technology (Hons)
 - CGPA : 3.18
- **UNIVERSITY COLLEGE TATI (UC TATI)** 2020 - 2021
 - Foundation of Engineering
 - CGPA : 3.07

Projects

- **Final Year Project - Studied Kinetics, Isotherms, and Thermodynamics of Methylene Blue Adsorption on Fly Ash from The Palm Oil Industry**
 - Conducted batch adsorption experiments (Kinetics, Isotherms, Thermodynamics).
 - Utilized UV-Vis for dye concentration analysis.
 - Optimized dosage and contact time for maximum adsorption efficiency.
- **Plant Design Project - Isopropanol Production Plant (40,000 MTA, 99.0% Purity)**
 - Simulated chemical processes using Aspen HYSYS.
 - Designed PFD (Process Flow Diagram) and P&ID (Piping & Instrumentation Diagram), including mass and energy balance calculations.
 - Estimated equipment sizing and calculated total capital cost.

Experience

- **QC Engineer - Japan Air Filter Asia** 08/2025 - Present
 - **Incoming Quality Control (IQC):** Handle IQC processes for all raw materials to ensure they meet specified standards.
 - **Supplier Coordination:** Liaise with suppliers regarding material quality issues and initiate Non-Conformance Reports (NCR) as needed.
 - **System & Documentation Management:** Maintain accurate records and update SAP and WMS for all material rejections.
 - **KIV Area Oversight:** Monitor and manage raw materials and finished goods placed in the KIV (Keep in View) area.
 - **Reporting:** Prepare and present weekly quality issue reports to the management team.
 - **Scrap Management:** Issue scrap forms for internally rejected materials.
 - **Artwork Verification:** Review and approve artwork related to incoming materials to ensure compliance with quality standards.
 - **Testing Procedures:** Conduct tensile strength tests for media rolls to validate material performance.
 - **Quality Assurance:** Ensure consistent quality of raw materials and finished goods throughout the production process.

Lab Services Intern - Sumisaujana TCM Chemicals Sdn. Bhd.

02/2025 - 07/2025

- Conducted QC testing on raw materials and finished emulsifier products using methods such as titration (acid/amine), iodine value, and specific gravity.
- Performed mud rheology and HTHP tests to evaluate product performance for oilfield drilling fluids.
- Blended graphene samples and supported analytical procedures related to chemical formulation.
- Documented and reported test results, supporting process improvements and product quality consistency.
- Followed strict laboratory safety protocols and environmental standards for chemical handling and disposal.

Skills

Technical Skills

- Chemical Process Design: Aspen HYSYS
- Process Control Systems: DCS
- Laboratory Techniques: UV-Vis spectroscopy, titration, chromatography
- Software: Microsoft Office Suite (Excel, Word, PowerPoint), AutoCAD
- Programming: Basic knowledge of C++ and Micro C
- Problem Solving and Data Analysis

Language

- Malay : Fluent
- English : Fluent

Reference

DR ABDUL HADI BIN ABDULLAH

Head of Programme

University College TATI (UC TATI), Kampus TATIUC, Jalan Panchur, Telok Kalong, 24000

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AP. Ts. DR. AHMED H.A DABWAN

FYP Supervisor

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